

LADC (Unit: 1, 2, 3)

ASSIGNMENT

- 1) Find the rank of the Matrices using Echelon form

$$\begin{bmatrix} 4 & 2 & -1 & 2 \\ 1 & -1 & 2 & 1 \\ 2 & 2 & -2 & 0 \end{bmatrix}$$

Ans: 2

$$\begin{bmatrix} 2 & -2 & 0 & 6 \\ 4 & 2 & 0 & 2 \\ 1 & -1 & 0 & 3 \\ 1 & -2 & 1 & 2 \end{bmatrix}$$

Ans: 3

- 2) Find the rank of the Matrices using normal form

$$\begin{bmatrix} 1 & -1 & 2 & 3 \\ 4 & 1 & 0 & 2 \\ 0 & 3 & 1 & 4 \\ 0 & 1 & 0 & 2 \end{bmatrix}$$

Ans: 4

$$\begin{bmatrix} 1 & 2 & 1 & 0 \\ -2 & 4 & 3 & 0 \\ 1 & 0 & 2 & -8 \end{bmatrix}$$

Ans: 3

- 3) Examine for consistency and if consistent then solve it.

$$\begin{cases} 2x_1 + x_2 - 5x_3 + x_4 = 8 \\ x_1 + 3x_2 - 6x_4 = -15 \\ 2x_2 - x_3 + 2x_4 = -5 \\ x_1 + 4x_2 - 7x_3 + 6x_4 = 0 \end{cases}$$

$$\begin{cases} x_1 = 3 \\ x_2 = -4 \\ x_3 = -1 \\ x_4 = 1 \end{cases}$$

$$\begin{cases} x_1 + x_2 + 2x_3 + x_4 = 5 \\ 2x_1 + 3x_2 - x_3 - 2x_4 = 2 \\ 4x_1 + 5x_2 + 3x_3 = 7 \end{cases}$$

Inconsistent

$$\begin{cases} 2x - y + 3z = 0 \\ 3x + 2y + z = 0 \\ x - 4y + 5z = 0 \end{cases}$$

$$\begin{cases} x = t \\ y = -t \\ z = t \end{cases}$$

- 4) Investigate for what values of a & b , the system of simultaneous equations

$$2x - y + 3z = 2$$

$$x + y + 2z = 2$$

$$5x - y + az = b$$

Have (1) No solution (2) A unique solution (3) An infinite number of solutions.

(Ans: 1) $a=8, b \neq 6$ (2) $a \neq 8, b \in R$ (3) $a=8, b=6$)

- 5) Examine for Linear dependence or independence the following system of vectors. If dependent, find the relation between them

Dependent

$$x_1 = (1, -1, 1), x_2 = (2, 1, 1), x_3 = (3, 0, 2)$$

$$x_1 + x_2 = x_3$$

$$x_1 = (3, 1, -4), x_2 = (2, 2, -3), x_3 = (0, -4, 1)$$

Dependent

$$2x_1 = 3x_2 + x_3$$

- 6) Find the Eigen values and corresponding Eigen vectors for the following matrices

I. $A = \begin{bmatrix} 9 & -1 & 9 \\ 3 & -1 & 3 \\ -7 & 1 & -7 \end{bmatrix}$

(Solution :- $-1, 0, 2$ and $\begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 4 \\ 1 \\ -3 \end{bmatrix}$)

II. $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 2 \\ 0 & 1 & 1 \end{bmatrix}$

(Solution :- $0, 2, -2$ and $\begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -3 \\ 1 \end{bmatrix}$)

III. $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$

(Solution :- $5, -3, -3$ and $\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}$)

- 7) Find A^4 with the help of Cayley Hamilton theorem

If $A = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$.

(Solution :- $\begin{bmatrix} -49 & -50 & -40 \\ 65 & 66 & 40 \\ 130 & 130 & 81 \end{bmatrix}$)

- 8) Solve by using Gauss elimination method

$$x + y - z = 2$$

$$2x + 3y + 5z = -3$$

$$3x + 2y - 3z = 6$$

Solution: $x = 1, y = 0, z = -1$

9) Solve by using Gauss Jordan method

$$2x + 4y + z = 3$$

$$3x + 2y - 2z = -2$$

$$x - y + z = 6$$

Solution: $x = 2, y = -1, z = 3$

10)

a) Solve by using Gauss Seidel method (Four Iterations)

$$10x + y + z = 12$$

$$2x + 10y + z = 13$$

$$2x + 2y + 10z = 14$$

Solution: $x = 1, y = 1, z = 1$

b) Solve by using Jacobi's iteration method (Four Iterations)

$$10x + y + z = 12$$

$$2x + 10y + z = 13$$

$$2x + 2y + 10z = 14$$

Solution: $x = 1, y = 1, z = 1$